🎬 Netflix Movies & TV Shows Data Analysis

A comprehensive Exploratory Data Analysis (EDA) project to analyze Netflix content trends, ratings, genres, and production patterns using Python.

📌 Project Summary

This project performs an in-depth Exploratory Data Analysis (EDA) on Netflix Movies and TV Shows datasets (titles.csv and credits.csv) to uncover meaningful insights about content distribution, ratings, genres, and industry trends.

The objective is to transform raw streaming data into actionable insights using data cleaning, visualization, and statistical analysis techniques. EDA helps identify patterns, trends, and relationships within the dataset.

🎯 Key Objectives

- Perform data cleaning and preprocessing
- Analyze Movies vs TV Shows distribution
- Study IMDB scores and popularity trends
- Identify top genres and production countries
- Analyze runtime and season distribution
- Explore actor and director contributions
- Generate visual insights for better understanding

🛠 Tech Stack

- Programming Language: Python
- Libraries Used: NumPy, Pandas, Matplotlib, Seaborn
- Environment: Jupyter Notebook

📂 Dataset Description

This project uses two datasets:

1. Titles Dataset (titles.csv)
- Contains information about Netflix content
- Columns: title, type, release_year, runtime, genres, country, IMDB score, TMDB score

2. Credits Dataset (credits.csv)
- Contains actor and director details
- Columns: name, role, character, person_id

🔎 Project Workflow

1️⃣ Data Loading
- Loaded datasets using Pandas
- Checked structure using head(), shape, columns

2️⃣ Data Cleaning & Preprocessing
- Handled missing and "unknown" values
- Converted data types (IMDB score, runtime, seasons)
- Extracted primary country and cleaned genres

3️⃣ Data Exploration
- Used info() and describe() for statistical summary
- Checked null values
- Analyzed categorical and numerical features

4️⃣ Data Visualization
- Movies vs TV Shows (Bar Chart)
- Age Certification Distribution
- IMDB Score Distribution (Histogram, Boxplot)
- Runtime Distribution
- Top Countries and Genres
- Content growth over years
- Actor vs Director analysis
- Top actors and directors
- Popularity vs Rating (Scatter Plot)
- TV show seasons distribution

📊 Key Insights

- Movies dominate Netflix content compared to TV Shows
- United States produces the highest content
- TV-MA and TV-14 are most common ratings
- Content increased significantly after 2015
- Drama and Comedy are most popular genres
- Most movies are 80–120 minutes long
- Most TV shows have 1–2 seasons
- IMDB scores mostly range between 6–8

📈 Conclusion

The analysis reveals that Netflix focuses heavily on movies and has expanded globally over time. The platform prioritizes diverse genres and mature-rated content to target a wide audience.

Drama remains the leading genre, while the United States dominates production. The project provides valuable insights into content strategy and audience preferences.